

# Tyson Gentry

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## SUMMARY

Computer Science student (expected December 2027) with strong foundations in algorithms, Python development, SQL, and Linux environments. Experienced in AI-assisted software engineering using modern LLMs such as GPT-5 and passionate about scalable AI systems and performance-focused development. Seeking a Summer 2026 Software Engineering Internship.

## EDUCATION

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| <b>University of Houston, College of Natural Sciences and Mathematics</b><br>Bachelor of Science in Computer Science,<br><b>Relevant Coursework:</b> Data Structures, Algorithms, Software Engineering, Database Systems, AI/ML | <b>Houston, TX</b><br>December 2027 |
| <b>Cedar Valley College, College of Science</b><br>Associates of Science                                                                                                                                                        | <b>Lancaster, TX</b><br>May 2024    |

## Technical Skills

- Languages: Python, C++, SQL, JavaScript, Swift
- Libraries/Frameworks: Pandas, NumPy, React
- Tools: Git, GitHub, Node.js, Linux, Excel, PowerPoint
- Concepts: Data Structures & Algorithms, Database systems, Computer Security, Computer Architecture

## Projects

### Shortest Path Graph Solver (C++)

- Engineered a high-performance graph solver using Dijkstra's algorithm in C++, optimizing for  $O(E \log V)$  time complexity using priority queues.
- Validated system accuracy by using a Linux-based testing suite to process and verify complex weighted graph inputs.

### Postal Service Database Project (MySQL, Node.js) — In Progress

- Collaborating on the design and development of a MySQL relational database for a postal service, including schema design, table creation, and structured data management.
- Developing a Node.js backend server to connect to the database, execute SQL queries, and support data access, retrieval, and update operations.

### Data Pipeline Automation (Python)

- Architected a data preprocessing pipeline that automated the cleaning and normalization of large-scale CSV datasets, reducing manual preparation time by an estimated 50%.
- Implemented data validation checks using NumPy to identify and handle null values and outliers, ensuring high-quality inputs for downstream analysis.
- Automated data preparation to improve usability for analysis workflows.

### Portfolio Website (React.js) — In Progress

- Developing a responsive React web application to showcase software projects, integrating AI tools (e.g., Claude, Cursor) for code refinement, architectural feedback, and debugging.
- Implementing interactive UI components and modern front-end practices while leveraging AI-assisted development to improve efficiency and optimize performance.

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| <b>Walmart – Associate</b>                                                                                                                                                                                                                                                                                                                                  | <b>Desoto, TX</b> |
| <ul style="list-style-type: none"><li>• Managed and audited inventory data for high-volume retail operations, ensuring 100% data integrity for thousands of SKUs via proprietary scanning software.</li><li>• Troubleshoot technical inconsistencies in scanning hardware/software, reducing processing delays and improving workflow efficiency.</li></ul> |                   |

## Leadership & Involvement

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|------------------------------------------------|----------------------|
| • National Society of Black Engineers – Member | (Aug 2024 – Present) |
| • Black Men in STEM                            | (Aug 2025 – Present) |
| • Collegiate 100 – Member                      | (Dec 2024 – Present) |
| • ColorStack – Member                          | (Aug 2025 – Present) |
| • Black Male Engineering Network               | (Dec 2024 – Present) |